

Domestic Strategy: Invest GBP1,000 at the 1.291 percent one-year British-pound risk-free rate to receive GBP1,012.99 ($= 1,000e^{(0.01291)}$) in one year.

Foreign Strategy: Convert GBP1,000 at USD/GBP1.2602 to receive USD1,260.20, which invested at the one-year US-dollar risk-free rate of 2.012 percent generates a return of USD1,285.81 ($= 1,260.20 e^{(0.02012)}$) in one year.

The no-arbitrage USD/GBP forward rate as of 31 May is therefore equal to 1.2693 ($= \text{USD1,285.81/GBP1,012.99}$).

As of 15 June:

Domestic Strategy: Invest GBP1,000 at the 1.562 percent one-year British-pound risk-free rate to receive GBP1,015.74 ($= 1,000e^{(0.01562)}$) in one year.

Foreign Strategy: Convert GBP1,000 at 1.2602 to receive USD1,260.20, which invested at the one-year US-dollar risk-free rate of 2.667 percent returns USD1,294.27 ($= 1,260.20 e^{(0.02667)}$) in one year.

The no-arbitrage USD/GBP forward rate as of 15 June is equal to 1.2742 ($= \text{USD1,294.26/GBP1,015.74}$).

As of mid-June, an investor agreeing to exchange US dollars for British pounds in one year must deliver more US dollars (1.2742 versus 1.2693) in exchange for GBP1 than if the same contract had been entered at the end of May. The greater increase in r_{USD} widens the interest rate differential ($r_{\text{USD}} - r_{\text{GBP}}$), causing US dollars to depreciate on a forward basis versus British pounds over the period. Differently put, the expectation for US-dollar depreciation on a forward basis versus British pounds would require a higher US-dollar interest rate to attract investors to US dollars versus British pounds.

Option Pricing Using Cash Flow Additivity

Let's assume an asset has a current price of 40 Chinese yuan (i.e., CNY40). The asset is risky in that its price may rise 40 percent to CNY56 during the next time period or its price may fall 20 percent to CNY32 during the next time period.

An investor wishes to sell a contract on the asset in which the buyer of the contract has the right, but not obligation, to buy the noted asset for CNY50 at the end of the next time period. We can apply the principle of cash flow additivity to establish no-arbitrage pricing for this contract.

The binomial tree in Exhibit 14 summarizes the two possible future outcomes of the asset: